

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	R KALAI AMUDHAN	Reg. No.:	192511292
Date:		Duration:	60 minutes

Code of Conduct

I R KALAI AMUDHAN (192511292) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R Kalai Amudhan

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

RESEARCH PAPER REVIEW / LITERATURE SURVEY

Problem Statement 17: Intelligent AI-Based Smart Dairy Farm Management System

Abstract

Modern dairy management is moving from periodic manual observation toward continuous, data-driven monitoring. The literature on precision livestock farming shows how sensors, connected devices, data platforms, and analytical models can convert observations of animal behaviour, environment, production, and health into actionable information. For the assigned smart-dairy problem, these technologies are relevant because a single management platform must combine heterogeneous streams rather than treating health, feeding, production, and environmental control as isolated activities. This survey reviews representative research contributions, compares their approaches, identifies limitations, and derives a practical research gap for an integrated AI-based system.

The analysis indicates that individual technologies are increasingly mature, but integration remains difficult. Sensor noise, missing data, animal-to-animal variation, limited labelled health datasets, model explainability, connectivity constraints, interoperability, cybersecurity, and the translation of predictions into farm-level actions remain important challenges. A promising direction is a modular architecture that performs edge-level preprocessing, cloud-based historical analytics, explainable AI inference, role-based dashboards, and continuous feedback from farm outcomes.

Keywords

Artificial Intelligence; Smart Dairy Farm; Precision Livestock Farming; IoT; Wearable Sensors; Machine Learning; Animal Health; Feeding Optimization; Milk Production; Predictive Analytics; Cloud Computing; Explainable AI.

Report Organization

The report first establishes the background and technology landscape, then reviews selected research papers individually. A comparative table is followed by critical analysis, research-gap identification, proposed system implications, future directions, and a consolidated conclusion. The structure follows the assessment requirements for paper selection, methodology review, comparison, limitations, research gaps, future work, and references.

1. INTRODUCTION AND PROBLEM CONTEXT

1.1 Background

Dairy farming involves continuous decisions concerning animal health, nutrition, reproduction, milking, environmental conditions, labour allocation, and record management. Traditional management can depend strongly on human observation and periodic measurement. Although experienced farmers can identify many visible changes, subtle changes in behaviour, rumination, activity, temperature, feed intake, or milk characteristics may be difficult to detect early. A smart dairy management system attempts to complement human expertise with continuous sensing and analytical support rather than replacing the farmer or veterinarian.

Precision livestock farming provides the broader technological context. The central concept is the use of automated measurements to observe individual animals or groups at a frequency and resolution that would be impractical through manual inspection alone. These measurements can be transformed into indicators of health, welfare, production, and environmental conditions. In software terms, the problem is therefore not only a machine-learning problem; it is a systems-engineering problem involving data acquisition, communication, storage, analytics, user interfaces, security, and maintainability.

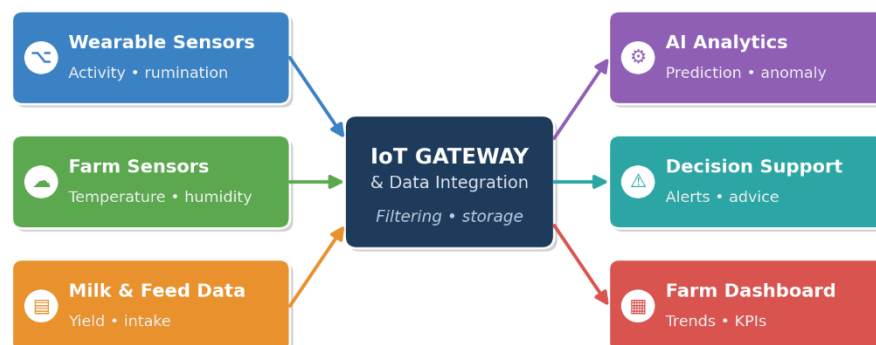
1.2 Relevance to the Assigned Problem

The assigned problem requires continuous cattle-health monitoring, disease prediction, feeding optimization, milk-production tracking, performance reporting, improved welfare, reduced operating cost, and increased productivity. These objectives create a multi-objective system. A health alert is useful only when it is timely and sufficiently trustworthy; a feeding recommendation is useful only when it considers animal condition and available resources; and a milk-production trend becomes more valuable when it can be related to health, nutrition, and environmental factors.

1.3 Major Technology Domains

- Wearable sensing captures activity, rumination, movement patterns, and other behavioural indicators.
- Environmental IoT measures temperature, humidity, air quality, and other barn-level conditions.
- Production monitoring records milk yield and, where available, quality-related measurements.
- Artificial intelligence supports classification, prediction, anomaly detection, and decision support.
- Cloud and edge computing provide scalable storage and analytics while allowing local preprocessing and low-latency responses.

Figure 1. Intelligent AI-Based Smart Dairy Farm Architecture



2. LITERATURE SELECTION AND REVIEW METHODOLOGY

2.1 Literature Survey Approach

The survey was organized around the assessment criteria supplied in the course document: relevant paper selection, review of objectives and methods, comparison of technologies and results, identification of limitations, research gaps, future directions, and consistent referencing. The course document specifically expects a comparative table containing author/year, research problem, methodology, datasets or tools, findings, limitations, and research gaps.

Representative scholarly works were selected from the established precision-livestock and dairy-monitoring literature. The selection emphasizes contributions to at least one major subsystem of the assigned problem: sensing, animal-health monitoring, precision dairy management, machine learning, data integration, or the wider precision-livestock framework. Because research papers use different experimental designs, direct numerical comparison is avoided where metrics are not measured under the same conditions.

2.2 Review Parameters

- Research objective: what practical or scientific problem the paper attempts to address.
- Methodology: sensing strategy, statistical method, machine-learning method, computer vision, or system architecture.
- Data and tools: type of observations, sensing devices, datasets, experimental setting, and software or analytical tools when reported.
- Outcome: the principal contribution or practical insight supported by the study.
- Limitation: constraints such as sample size, sensor dependence, environment, labels, generalization, cost, or integration.
- Relevance: how the contribution maps to the intelligent smart-dairy problem.

2.3 Key Evaluation Perspective

A useful literature survey should not simply list papers. It should identify relationships among them. Sensor-focused research establishes how animal information can be measured; analytics-focused research establishes how patterns can be inferred; precision-livestock research explains how measurements can support management; and software-oriented design connects these capabilities into a dependable platform. The assigned system requires all four perspectives.

2.4 Important Caution in Comparing Studies

Accuracy values reported in different papers should not be treated as interchangeable. A model tested on a controlled dataset may perform differently from the same model on a different farm, breed, season, sensor device, or disease prevalence. Therefore, this report uses qualitative comparison for cross-paper analysis and treats reported performance as study-specific evidence rather than a universal benchmark.

3. INDIVIDUAL RESEARCH PAPER REVIEWS – PART I

3.1 Berckmans (2017) – General Introduction to Precision Livestock Farming

Berckmans presents precision livestock farming as an approach in which automated monitoring and control technologies are used to support livestock production and management. The paper is valuable for the assigned problem because it establishes the conceptual foundation for transforming continuous measurements into management information. Rather than viewing sensors as isolated devices, the precision-livestock perspective emphasizes the relationship between measurement, interpretation, and intervention.

The major methodological contribution is conceptual and systems-oriented. It describes how automated monitoring can observe animals continuously and how deviations from expected patterns can support early detection and management. For the smart dairy system, this perspective supports the use of animal-level sensing combined with analytics and decision support.

A key limitation is that a broad conceptual framework does not by itself solve the practical issues of sensor calibration, disease labels, model training, interoperability, or deployment cost. The paper therefore provides system-level motivation but must be complemented by empirical studies and machine-learning research.

3.2 Rutten et al. (2013) – Sensors to Support Health Management on Dairy Farms

Rutten and colleagues review sensor technologies intended to support health management in dairy farming. The work is directly relevant to the health-monitoring requirement because it examines how sensor-derived information can assist the detection or management of health-related conditions. The study demonstrates the importance of selecting measurements that have a meaningful relationship with biological changes rather than collecting data simply because a sensor is available.

The reviewed approaches include automated measurements associated with animal behaviour, physiology, production, and related farm observations. This provides a useful design lesson: a smart dairy platform should treat sensing as a pipeline from measurement to validated indicator and finally to an actionable alert.

3.3 Neethirajan (2020) – Role of Sensors, Big Data and Machine Learning in Modern Animal Farming

Neethirajan discusses the growing role of sensors, large-scale data processing, and machine learning in modern animal farming. The paper is important because it connects data collection with computational intelligence. In the assigned smart-dairy system, this connection supports a layered architecture in which sensor data are collected continuously, transformed into useful features, analysed by predictive models, and presented through a decision-support interface.

The paper also reinforces the importance of big-data characteristics such as volume, velocity, variety, and the need for meaningful interpretation. For a dairy platform, data may arrive from wearables, environmental sensors, milk systems, feeding records, and manual observations. A scalable data architecture is therefore essential.

4. INDIVIDUAL RESEARCH PAPER REVIEWS – PART II

4.1 Banhazi et al. (2012) – Precision Livestock Farming: An International Review

Banhazi and colleagues discuss precision livestock farming from an international perspective, including technological development and the practical movement toward automated measurement and management. The work is relevant because it places smart-farm technologies within a broader production system rather than treating them as independent innovations.

The contribution supports the idea that a successful smart farm requires integration of hardware, information systems, management practices, and user decisions. This is directly applicable to the assigned dairy system, where a disease prediction model without a usable alert mechanism would have limited operational value.

A limitation is the changing technology environment. Sensor capabilities, communication technologies, cloud platforms, and machine-learning methods have evolved substantially since the paper was published. Consequently, the conceptual lessons remain useful, but implementation decisions should use contemporary architectures and current validation practices.

4.2 Bewley (2010) – Precision Dairy Farming: Applications, Outcomes and Future Prospects

Bewley's work on precision dairy farming discusses how technology can support dairy-management decisions through automated measurement and analysis. The relevance to the assigned problem lies in the connection between data and practical farm outcomes. Production, health, reproduction, and management data can be used together to support more timely decisions.

The paper's systems perspective helps define a key requirement for the proposed solution: the platform should provide information at the level at which a decision is made. Farmers do not simply need raw sensor values; they need interpretable indicators, trends, warnings, and recommended next actions.

4.3 Halachmi et al. (2019) – Smart Animal Agriculture

Halachmi and colleagues discuss smart animal agriculture and the use of real-time sensors to improve animal well-being and production. This literature is particularly relevant to the proposed system because it frames precision monitoring as an interaction among sensing, analytics, management, and animal welfare.

For the assigned problem, the implication is that the platform should not optimize productivity alone. Health and welfare indicators must be treated as first-class system objectives. A recommendation that increases short-term production but causes sustained welfare deterioration would not satisfy the broader goals of responsible dairy management.

4.4 Synthesis of the Reviewed Papers

Across these studies, a consistent theme emerges: smart livestock systems are strongest when measurement, analytics, and management action are designed as one pipeline. The literature also shows why single-sensor or single-model solutions can be fragile. Farm conditions vary, biological signals are noisy, and the meaning of an observation depends on animal history and context. These observations motivate the multi-modal, explainable, and feedback-oriented architecture proposed later in this report.

5. TECHNOLOGY-WISE ANALYSIS

5.1 Wearable and Animal-Level Sensing

Wearable devices can provide continuous measurements that are difficult to obtain manually at the same frequency. Activity, movement patterns, rumination-related behaviour, and other signals can serve as indicators of changes in animal condition. For the proposed platform, wearable data should be associated with an individual animal identifier and timestamp so that changes can be interpreted against the animal's history.

The main software challenge is signal quality. Devices may lose connectivity, batteries may discharge, sensors may shift position, and observations may contain outliers. A robust platform should therefore include validation rules, missing-data handling, calibration records, and confidence scores.

5.2 Environmental IoT

Environmental conditions influence animal comfort and farm operations. Temperature and humidity measurements can be combined into environmental indicators, while additional sensing can support ventilation and air-quality monitoring. The value of these data increases when they are linked to animal observations. For example, a change in activity coinciding with an environmental change may require a different interpretation from an isolated activity change.

5.3 Milk Production and Quality Monitoring

Milk production is both an outcome variable and a source of information. A sudden change in yield can indicate nutritional, health, reproductive, equipment, or management factors. The system should therefore maintain historical production records and support trend analysis rather than displaying only the latest value.

5.4 Machine Learning and Predictive Analytics

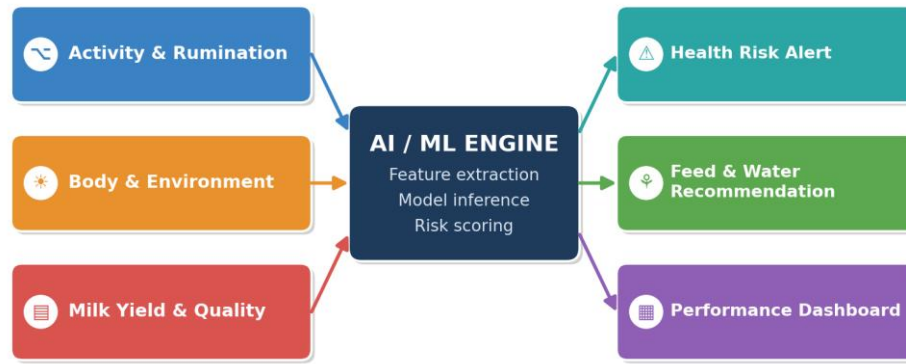
Machine learning can support classification, regression, anomaly detection, risk scoring, and time-series forecasting. Disease prediction can be formulated as a risk-estimation problem in which the model uses behavioural, physiological, environmental, and production features. Feeding optimization can use a combination of animal requirements, production level, available feed, and historical response.

Model selection should be driven by the operational requirement. Complex deep-learning models may capture nonlinear relationships but can require more data and computational resources. Simpler models can be easier to validate and explain. The most appropriate system may therefore use different models for different tasks rather than forcing one algorithm to solve every problem.

5.5 Edge, Cloud and Data Management

Edge processing can reduce latency and bandwidth requirements by filtering or aggregating sensor data near the farm. Cloud services can provide long-term storage, fleet-level analytics, dashboards, model training, and centralized updates. A hybrid architecture can combine both: urgent local alerts can be generated close to the data source, while historical analysis and model retraining can occur centrally.

Figure 2. AI Decision Pipeline for Dairy Farm Management



6. COMPARATIVE LITERATURE SURVEY

The following table synthesizes the reviewed literature using the parameters requested by the assessment. The comparison is qualitative because the papers differ in objectives, datasets, experimental settings, and evaluation metrics.

Author / Year	Research Problem	Methodology / Approach	Data / Tools	Key Contribution	Limitation	Research Gap
Berckmans (2017)	Continuous livestock monitoring	Precision livestock farming framework	Automated animal/farm measurements	Defines measurement-to-decision concept	Broad conceptual scope	Need integrated, validated farm platforms
Rutten et al. (2013)	Sensor-supported dairy health management	Review of health-related sensor applications	Animal behaviour, physiology and production sensors	Shows potential of sensors for health support	Signal quality and validation remain difficult	Multi-modal, farm-robust health models
Neethirajan (2020)	Digital transformation in animal farming	Sensors + big data + ML perspective	IoT data and computational analytics	Connects sensing with intelligent analysis	General review; deployment context varies	Operationally validated AI pipelines
Banhazi et al. (2012)	Precision livestock adoption	International PLF review	Automated monitoring and management technologies	Highlights integration and adoption	Technology has evolved	Interoperability and scalable implementation
Bewley (2010)	Technology-enabled dairy decisions	Precision dairy management perspective	Production and management data	Connects data with farm decisions	Adoption, cost and complexity	User-centred decision support
Halachmi et al. (2019)	Real-time smart animal management	Sensor-enabled smart agriculture	Real-time animal and farm measurements	Links technology with welfare and production	Requires practical validation	Responsible, welfare-aware automation

6.1 Comparative Interpretation

The comparison shows that the literature is distributed across layers of the smart-dairy problem. Some papers concentrate on the measurement layer, others on management or analytical concepts. This distribution is useful because it reveals that the proposed system is an integration challenge. The research gap is not simply the absence of another classifier; it is the need to combine heterogeneous sensing, reliable data engineering, explainable prediction, decision support, and continuous validation in a maintainable platform.

6.2 Parameters for Evaluating the Proposed System

- Health-prediction quality: precision, recall, sensitivity, specificity, calibration, and alert timeliness.
- Operational performance: response time, data-processing throughput, availability, and energy consumption.
- Scalability: ability to support more animals, sensors, barns, and historical records.
- Usability: clarity of alerts, dashboard readability, workflow fit, and actionability.
- Robustness: tolerance to missing, noisy, delayed, or inconsistent sensor data.
- Responsible AI: explainability, audit trails, security, privacy, and controlled model updates.

7. CRITICAL ANALYSIS, LIMITATIONS AND RESEARCH GAP

7.1 Common Limitations in Existing Approaches

A major limitation across smart-livestock research is data heterogeneity. Different farms use different devices, sampling frequencies, identifiers, communication protocols, and management practices. Even when two studies measure the same concept, their raw signals may not be directly comparable. A software platform must therefore include a normalization and metadata layer.

Another limitation is the availability of high-quality labels. Disease prediction requires trustworthy examples of healthy and affected states, including information about when a condition begins and when it is clinically confirmed. If labels are delayed or inconsistent, a model can learn correlations that appear accurate during testing but fail during real deployment.

Generalization is also difficult. Animal breed, age, lactation stage, housing, climate, feed, herd size, and management practices can influence observed patterns. A model trained on one farm may therefore experience distribution shift when moved to another farm. Continuous validation and farm-specific calibration are necessary.

Interpretability is a further challenge. A farmer or veterinarian may hesitate to act on a high-risk score if the system cannot explain the evidence behind the alert. Explainable features, confidence indicators, trend views, and event histories can improve trust.

7.2 Identified Research Gap

The central research gap identified from the reviewed literature is the limited integration of heterogeneous dairy-farm data into a single, explainable, continuously learning decision-support system. Existing research demonstrates the value of sensors, precision livestock principles, machine learning, and automated monitoring, but these capabilities are often studied as separate components. A practical platform must connect them while maintaining data quality, security, interoperability, and human oversight.

7.3 Proposed Research Gap Statement

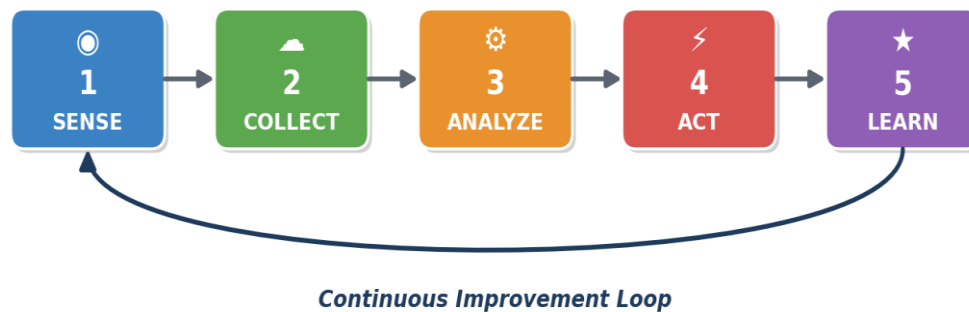
There is a need for an integrated AI-based smart dairy management architecture that combines wearable, environmental, production, feeding, and management data; performs robust preprocessing and multi-modal analytics; produces explainable health and productivity insights; supports real-time alerts and recommendations; and continuously evaluates model performance under changing farm conditions.

7.4 Why the Gap Matters

Closing this gap can improve the usefulness of AI from a demonstration technology into an operational management tool. The value is not only higher predictive performance. The system must reduce unnecessary alerts, support earlier intervention, help allocate labour, improve resource planning, and provide evidence for management decisions while keeping animal welfare central.

Figure 3. Closed-Loop Smart Dairy Management Cycle

Sensor data → decisions → outcomes → new data → model refinement



8. PROPOSED SYSTEM IMPLICATIONS AND FUTURE RESEARCH DIRECTIONS

8.1 Proposed Modular Architecture

Based on the literature analysis, the proposed system can be organized into six modules: sensing, communication, data management, AI analytics, decision support, and reporting. Modular design is important from a software-engineering perspective because individual components can evolve without rewriting the complete platform.

- Sensing module: receives wearable, environmental, production, and feeding observations.
- Data-ingestion module: validates timestamps, animal identifiers, ranges, units, and communication status.
- Analytics module: extracts features, detects anomalies, predicts risk, and estimates trends.
- Decision-support module: converts model output into prioritized alerts and recommendations.
- Dashboard module: presents animal-level and herd-level views with history and drill-down.
- Administration module: manages users, devices, thresholds, model versions, audit logs, and configuration.

8.2 Multi-Modal Disease Prediction

Future work should investigate models that combine multiple evidence streams rather than relying on a single sensor. A multi-modal approach can incorporate activity, rumination, body-related measurements, environmental conditions, milk-production trends, and historical events. Fusion may occur at the feature level, model level, or decision level. The selected strategy should be evaluated for robustness when one data stream is unavailable.

8.3 Explainable AI

Explainability should be treated as a system requirement rather than a decorative feature. An alert can include the observed trend, contributing indicators, confidence level, comparison with the animal's baseline, and recommended verification steps. Such explanations can help the user distinguish between a strong warning and a weak anomaly requiring observation.

8.4 Continuous and Federated Learning

As farms accumulate data, models can be updated to reflect changing conditions. Future research can explore continual learning while preventing catastrophic forgetting and maintaining auditability. Federated learning may also be investigated where data should remain at farm sites while model knowledge is aggregated across locations. These approaches require careful evaluation of privacy, communication overhead, model drift, and security.

8.5 Digital Twin and Simulation

A digital representation of the herd and farm environment could support simulation of feeding, environmental, and production scenarios. A digital-twin approach can help evaluate recommendations before they are applied, especially for decisions that have cost or welfare implications. Such simulation should complement—not replace—real-world validation.

8.6 Human-in-the-Loop Management

The proposed system should keep humans in the decision loop for clinically significant interventions. AI can prioritize animals and provide evidence, while farmers and veterinarians can confirm observations and record outcomes. These feedback records can become valuable labelled data for future model improvement.

9. RESPONSIBLE SOFTWARE ENGINEERING AND SOCIETAL IMPACT

9.1 Animal Welfare

A smart dairy platform should evaluate success using welfare indicators as well as production indicators. Continuous monitoring can support earlier recognition of abnormal behaviour and environmental stress, but automated alerts must be designed to avoid alarm fatigue. A system that generates too many low-value warnings can reduce attention to important events.

9.2 Data Privacy and Security

Although animal data are not equivalent to personal medical data, farm records can still be commercially sensitive. The platform should use authentication, authorization, encryption, secure device communication, audit logging, and controlled data sharing. User roles should reflect operational responsibilities so that only authorized personnel can change system configuration or access sensitive records.

9.3 Model Governance

Every deployed model should have a version, training-data description, evaluation record, deployment date, and rollback mechanism. Monitoring should detect changes in input distributions and performance. When a model is updated, the system should preserve enough history to determine which version produced a particular alert.

9.4 Interoperability and Maintainability

A farm may use devices from multiple vendors. Vendor-neutral interfaces, documented APIs, standard data schemas, and device abstraction can reduce lock-in and make future expansion easier. Maintainable software should separate business rules from device-specific code and should support automated testing, logging, monitoring, and controlled deployment.

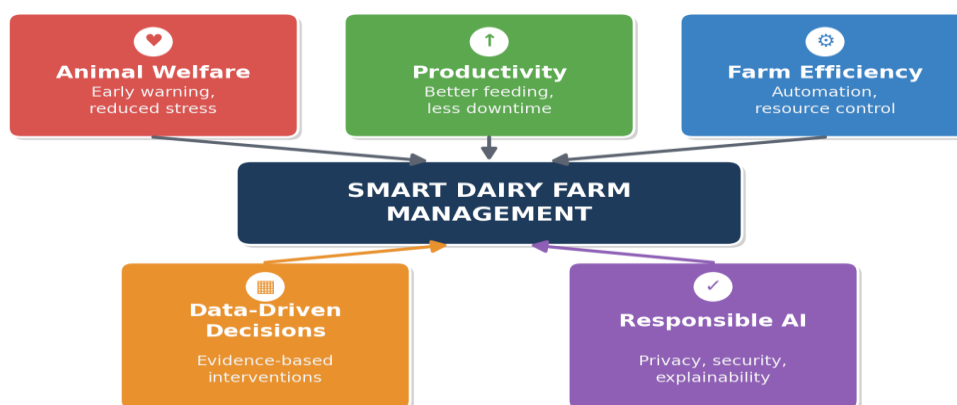
9.5 Economic and Workforce Impact

Automation can reduce repetitive monitoring and help allocate labour to higher-value tasks, but it also changes the skills required on farms. Users may need training in dashboard interpretation, sensor maintenance, data quality, and AI-assisted decision making. Cost-benefit analysis should therefore include equipment, connectivity, maintenance, training, and replacement costs rather than considering model accuracy alone.

9.6 Environmental Sustainability

Better monitoring may support more efficient use of feed, water, energy, and labour. However, digital systems also consume electricity and create hardware waste. Sustainable design should consider low-power sensing, efficient data transmission, device repairability, lifecycle management, and responsible disposal of electronic components.

Figure 4. Expected Impact Areas of an AI-Based Smart Dairy System



10. CONCLUSION

The literature survey demonstrates that intelligent dairy management is supported by a strong foundation of precision livestock farming, sensor-based monitoring, automated data collection, machine learning, and digital decision support. The reviewed studies collectively show that continuous measurement can reveal information that is difficult to obtain through occasional manual observation. For the assigned problem, this evidence supports the integration of animal health monitoring, disease-risk prediction, feeding support, milk-production tracking, environmental monitoring, and farm analytics.

At the same time, the literature makes clear that simply adding more sensors or choosing a more complex machine-learning algorithm does not automatically create an effective smart-farm system. Real deployment must address missing and noisy data, differences between farms, limited labelled health events, model drift, interoperability, connectivity, cybersecurity, explainability, user acceptance, and cost. These issues are software-engineering concerns as much as they are agricultural or artificial-intelligence concerns.

The most significant research gap identified in this survey is the need for an integrated and explainable platform that connects heterogeneous data sources to actionable decisions while preserving human oversight. The proposed architecture addresses this gap through modular sensing, data validation, AI analytics, decision support, dashboards, and continuous feedback. A closed-loop approach can allow the system to learn from farm outcomes while maintaining traceability and responsible model governance.

Future research should validate the architecture using multi-farm datasets and real operational workflows. Evaluation should include predictive quality, alert timeliness, robustness to missing data, scalability, usability, welfare outcomes, and economic impact. If these dimensions are evaluated together, the resulting system can move beyond a laboratory prototype toward a dependable software platform capable of supporting modern dairy-farm management.

11. REFERENCES

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